

1 Introduction: An Ancient Sprinter

In 2003, Mary Pappen Jr., a young Mutthi Mutthi Aboriginal Australian girl, discovered footprints in the ground of an old, dried-up lakebed.



A scientist named Steve Webb studied the footprints. He claimed some of them were made by a person running barefoot through the mud as fast as a modern day sprinter in the Olympics. News outlets picked up the story, and soon people were talking about the ancient sprinter who could rival today's best. However, a few years later, the story was essentially forgotten.

How did Webb estimate the speed of this ancient sprinter? How reliable was his work, and why did the story seem to die away in just a few years? That's what we'll examine in this handout as we learn about speed and motion.

2 From Footprints to Dot Diagrams

After Pappen's discovery, people studied the footprints in different ways. For example, Webb used a high-tech method called "optically stimulated luminescence" to estimate that the footprints were between 19,000 and 23,000 years old. This method uses specialized, sensitive equipment and is based on the theories of radioactive decay and how electrons move in the small grains of minerals found in dirt.

A group of Pintubi Aboriginal Australian trackers came to study the footprints as well. Using their lifelong experience with human footprints, they realized some of the tracks were made by families walking, but a few were made by a group of hunters chasing an animal. One hunter in particular was running fast to cut off the animal's escape. This hunter was the ancient sprinter from Webb's report.

To estimate how fast the hunter was running, Webb used neither high tech equipment nor expert judgment. Instead, he made measurements with a tape measure and then used a *conceptual model*. A conceptual model is a simplified way of thinking about a specific part of the world. Conceptual models can combine measurements, equations, diagrams or pictures, graphs, and even computer programs to explain something. They are never perfect, but they can still be good enough to help us learn new things.

To understand the model Webb used, we can begin with a diagram of the footprints left by the sprinting hunter. Webb measured the stride length, which he defined to be the distance between two footfalls of the same foot, so that one stride was two steps. He also measured the foot length.



Let's assume the hunter was taking 100 strides per minute, so that time between strides was $\frac{60 \text{ s}}{100} = 0.6 \text{ s}$ apart. Then the diagram shows where the right foot of the hunter was once every 0.6 s. We can leave the left foot out; it isn't needed in this model.

We can simplify even more by replacing the footprint with a dot. Here is a diagram with those two simplifications. We're also zooming out to show six strides now. (The previous diagram showed only two.)



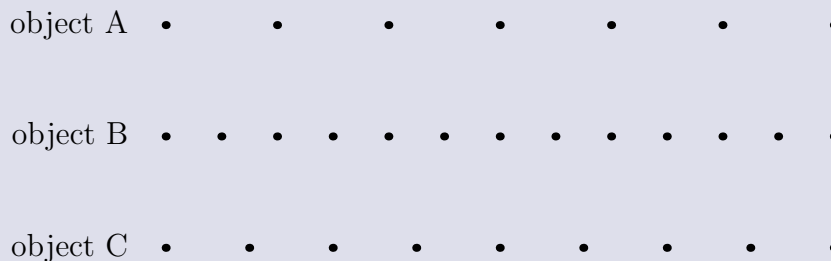
This sort of picture, using dots spaced an equal time apart to represent motion, is called a *dot diagram*. It's very simplified, but it represents the core information: the stride length.

A dot diagram can model anything that moves. For example, imagine a car dripping a drop of wet paint every 0.6 s. The dots of paint on the road would form a dot diagram.

There is no need for the dots to be drawn every 0.6 s. Within a dot diagram, we need the same time between all the dots, but that time can be any amount of time we wish. However, if we want to compare two different dot diagrams, they should have the same time between dots.

Additionally, we can use dot diagrams even when there are no actual dots. We can take any object and imagine it leaving dots behind as it moves.

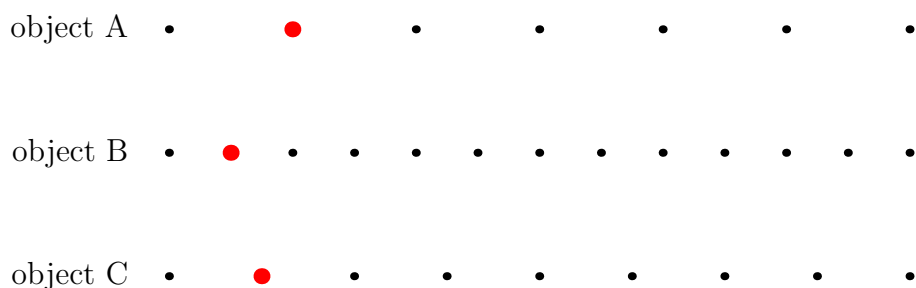
Problem 2.1. Below are dot diagrams for three objects, each moving a different speed. The time between dots is the same for each object. Rank the objects from fastest to slowest.



Solution: All the dots diagrams have the same time between dots. Object B's dots are close together, but this is not because the dots are being drawn very quickly. The dots are being drawn at the same rate for object B as for objects A and C, so the spacing difference must be due to speed differences between the objects.

We can imagine the first dots in each row being laid simultaneously, as if all three objects were starting a race together.

Each row would have its second dot laid at the same time as well. We can highlight those second dots in red to visualize them:



Whichever red dot is furthest along is the object that moved the most at this point in the race. The further an object has moved, the faster it must be going.

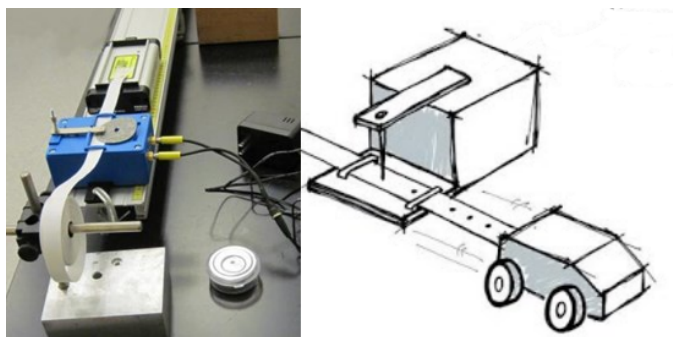
Based on which red dot is furthest ahead, the ordering of speeds is $A > C > B$.

Takeaway: the further apart the dots, the faster the object was going. This is because a fast-moving object can travel further than a slow-moving one in the time between dots being laid down. $\square$

Exercises

- 2.1. Why do you think Webb measured the stride length of the hunter, rather than the step length? Remember that a stride is between two prints of the same foot, and a step is between one left footprint and one right footprint.
- 2.2. In the example problem, do the dot diagrams represent the motion of each object for the same amount of time, or do some show the object's motion over a longer time than others? Explain how you know.
- 2.3. How long did it take the hunter to cover the distance in the dot diagram with six dots spaced 0.6 s apart?
- 2.4. Does a dot diagram tell you which direction the object was moving? Why or why not?
- 2.5. Draw a dot diagram for a ball rolling slowly across the floor, then for one rolling quickly.
- 2.6. You are given a dot diagram and can use a ruler to measure the distance between the dots. What other information do you need to know before you can find the speed of the object that the dot diagram represents?
- 2.7. One way to generate a dot diagram for a cart is to attach a long, thin strip of paper behind the cart. The cart drags the paper forward as it goes. The paper is pulled underneath a "ticker", a device with a sharp needle that

bounces up and down at a regular rate, punching a hole in the paper each time it bounces. These holes function as the dots in a dot diagram. A photo and diagram of this setup are shown below:



You adjust the ticker to tick more slowly, so there is a longer time between it making dots in the paper. Would this increase or decrease the spacing between the dots?

2.8. You want to use the ticker setup described above on a fast car and a slow car, then use the dot diagrams generated to compare the cars' speeds. Should you set the ticker to always tick at the same rate for both cars, or should it tick faster for the fast car and slower for the slow car, or should it tick faster for the slow car and slower for the fast car?

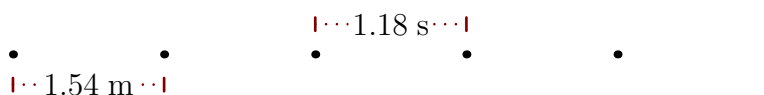
2.9. Imagine you want to use the ticker setup to draw a dot diagram for a car that gets faster and faster over time. Should you set the ticker to always tick at the same rate, to tick faster and faster over time, or to tick slower and slower over time?

3 Speed: Distance and Time

We've modeled footprints with a dot diagram, and now we want to incorporate speed into the model. To see how that can work, let's start with a problem.

Problem 3.1. While walking at a steady pace, you take one stride every 1.18 s. Your stride length is 1.54 m. (The symbol s is for seconds and m is for meters.) How fast are you going in meters per second?

Solution: We begin with a dot diagram.



Before doing a calculation, let's think about the relationship between speed, distance, and time using a simple example.

You travel $30 \frac{\text{mi}}{\text{h}}$ for 2 h. How far do you go? (The mi is miles and h is for hours.)

Both traveling for a long time and traveling at high speed result in going further. To find the distance traveled, we multiply the speed by the time, getting $\left(30 \frac{\text{mi}}{\text{h}}\right) \cdot (2 \text{ h}) = 60 \text{ mi}$.

As an equation,

$$\text{distance} = \text{speed} \cdot \text{time}. \quad (3.1)$$

We want speed, so we can algebraically solve for it, getting

$$\text{speed} = \frac{\text{distance}}{\text{time}}. \quad (3.2)$$

This equation says that for a fixed time, the bigger the distance, the faster the speed. For example, imagine two objects. The first goes 10 m in 2 s and the second goes 1000 m in 2 s. The second object had greater speed.

Also, making the denominator of a fraction smaller makes the entire fraction bigger. The interpretation here is that for a fixed distance, the less time it takes you to travel that distance (smaller denominator), the greater your speed (larger entire fraction).

For example, imagine one object travels 10 m in 2 s and another object travels 10 m in just 0.02 s. The second object was traveling at much higher speed.

Now we're ready to use this equation with the values in the problem. The distance you travel with each stride is 1.54 m and the time it takes is 1.18 s, so your speed is

$$\text{speed} = \frac{1.54 \text{ m}}{1.18 \text{ s}} = \boxed{1.31 \frac{\text{m}}{\text{s}}}. \quad (3.3)$$

This is about 3 mi/h, meaning it would take about 20 minutes to walk one mile at this speed, or about 13 minutes to walk one kilometer. That's a normal walking speed for most people. $\square$

Although we didn't use the dot diagram directly in calculating the speed, drawing and labeling it can help organize our thoughts before we do the calculation.

In this problem, we had to be given the time between strides in order to find the speed. If the problem had only said that our strides were 1.54 m long but not told us the time between strides, we could have drawn a dot diagram, but we couldn't find the speed because we wouldn't understand how quickly we were taking those strides.

Webb measured the ancient hunter's stride length, but not how fast the strides were taken. He could have tried to guess the time between strides, but there's a better method than pure guessing.

Exercises

3.1. A dot diagram has dots spaced 0.05 m apart and the time between dots is 10 s. Suggest an animal that typically moves at about the speed of the object modeled by this dot diagram.

3.2. You are traveling at 3.2 m/s with a stride length of 2.08 m. Find the number of seconds between each stride you take.

3.3. A car is traveling at 25 m/s down a freeway. There is some wet paint on one of the wheels, so every time the wheel goes around, it leaves a dot. The wheel goes around once every 0.079 s. Find the distance between the dots, and interpret what this distance means, i.e. where could it be measured on the car?

- 3.4. Find two different combinations of distance between dots on a dot diagram and time between dots on a dot diagram that yield the same speed for the object being modeled. You can make up any numbers you want.
- 3.5. You have two dot diagrams. The dots are twice as far apart on diagram A as on diagram B. However, the time between recording dots was three times as high for diagram A as it was for diagram B. Which object was moving faster?
- 3.6. In the problem above, object A was moving at 10 m/s. How fast was object B moving?

4 A model of stride time

Although Webb had to estimate the time between strides, he didn't have to guess blindly.

Imagine a child and an adult walking together. The child is short, so to keep up with the adult, they have to take more steps than the adult, and the time between their steps will be smaller than for the adult. This suggests that height affects time between strides.

Webb didn't know how tall the ancient hunter was, but he did have one useful piece of information: the foot length. Tall people tend to have longer feet, so Webb could use the foot length to guess the hunter's height. Then he could combine the height and stride length to guess the time between the hunter's strides, and finally use stride length and time between strides to calculate the speed.

Webb didn't do all that work on his own. Instead, he relied on a model built decades earlier by three Canadian scientists. The Canadian team used data on modern day humans to create a series of equations that take as inputs the stride length and foot length and give an estimated time between strides as output. These equations, along with some plots they made of the same information, formed their model of human movement.

When Webb used the Canadian scientists' equations, he found a time of 0.360 s between strides for the hunter. He also measured a stride length of 3.73 m. Combining these, he found a speed of $\frac{3.73 \text{ m}}{0.360 \text{ s}} = 10.36 \text{ m/s}$.

Problem 4.1. Compare the running speed Webb estimated for the ancient hunter to Usain Bolt's running speed. Bolt holds the world record for the 100 m with a time of 9.58 s.

Solution: Bolt's average speed over the entire race was

$$\frac{100 \text{ m}}{9.58 \text{ s}} = 10.44 \text{ m/s.} \quad (4.1)$$

According to this calculation, Usain Bolt was just a little bit faster than the ancient hunter, but they ran at almost the same speed.

□

That summarizes how Webb estimated the hunter's speed, but we might wonder: how reliable are those results?

Exercises

- 4.1. Suggest three different sources of uncertainty in Webb's calculation.
- 4.2. Suggest a reason that the calculation we've done here doesn't give Usain Bolt's fastest possible sprinting speed.

5 Questioning a model

If we want to know whether the model Webb used gives reasonable results, we could look at some of its claims besides the speed of the hunter, and see how reasonable they are.

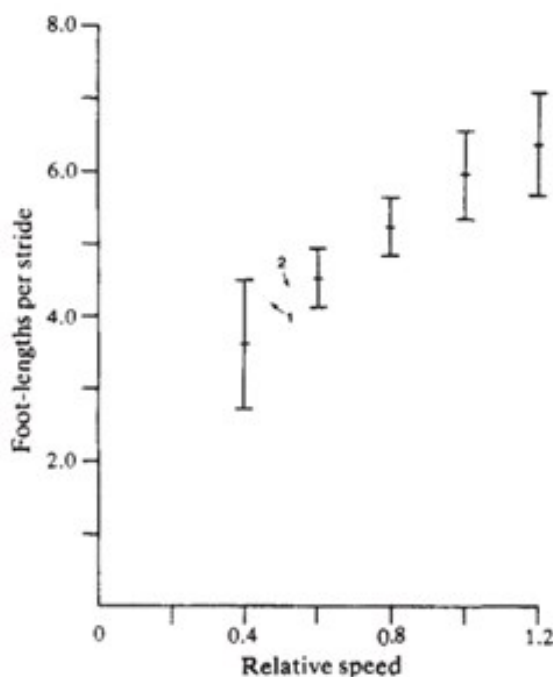
When we do this, things don't look great for the model. Usain Bolt's average stride length in his world record was 4.88 m, much longer than the 3.73 m stride length of the ancient hunter. With only that information, it would make sense to guess that Bolt was running significantly faster. It's only because the model predicted an extremely short time between strides for the hunter (0.360 s for the hunter as opposed to 0.467 s for Bolt) that the hunter came out around the same speed.

Perhaps that's because the hunter was much shorter than Bolt and short people take less time between strides? Usain Bolt is 195 cm tall, and the model predicted that the hunter was 194 cm tall - essentially the same height.

Webb applied the same model to several other tracks near the site. None of them came out to have running speeds as fast as the hunter, but there were many other unusual numbers, including many instances of very short times between the strides.

When we question our models and test how reasonable their outputs are, we're doing a *sanity check*. In this case, the model Webb used seems a bit unbelievable, and doesn't quite pass our sanity checks.

When we dig deeper into the model, we can see what happened. Here is a plot that the Canadian scientists made showing the experimental data they were modeling:



This plot shows *relative speed* on the horizontal axis. This is defined as speed divided by height. In other words, your relative speed is how many times your own body length you run each second.

Usain Bolt's relative speed was $\frac{10.44 \text{ m/s}}{1.95 \text{ m}} = 5.35 \frac{1}{\text{s}}$.

As we see from the chart, the fastest relative speed used in making the model was the rightmost data point, which had a relative speed of about $1.2 \frac{1}{\text{s}}$.

The equations Webb used were made to model humans walking, not running. When Webb took the equations from the Canadians' model and applied them to the hunter, he was able to calculate something, but the calculation wasn't meaningful. It used inputs from beyond the region where the model was intended to work.

If Steve Webb had understood the Canadians' model better before using it, he could have realized the model wasn't appropriate for estimating the speed of the hunter. Or, after using the model, he could have done some sanity checks and noticed that the results were unrealistic. He could also have asked a colleague with more experience in using these models for help. Unfortunately, neither Steve Webb, nor any of the scientists who reviewed his work, noticed the mistake.

From there, the inappropriately calculated speed spread to news stories and books. Journalists and authors repeated the figure without doing a critical check on where it came from.

In 2013, Javier Ruiz and Angélica Torices, two Spanish biologists and paleontologists, re-analyzed the hunter's footprints using a new model that was designed for running speeds. Their model was tested against real data from people running. Ruiz and Torices estimated a running speed of $7.2 \pm 1.0 \text{ m/s}$ for the hunter. This means that they couldn't tell exactly how fast he was going, but estimated that his speed was probably between 6.2 m/s and 8.2 m/s . The hunter, while running quite fast, was not running nearly as fast as modern Olympic sprinters.

This is why when we generate models, we need to test them against real data to determine their reliability. We also want to remember the limitations of our models and do sanity checks on our results. Finally, we should report our results with uncertainty whenever we can to indicate just how accurate we think our findings are.

Exercises

- 5.1. Calculate the percentage uncertainty implied by Ruiz and Torices's estimate of $7.2 \pm 1.0 \text{ m/s}$ for the hunter's tracks.
- 5.2. A child and an adult are walking along together at the same speed. Which has the greater relative speed?
- 5.3. What was the time between strides that Ruiz and Torices found for the hunter? What was their uncertainty on this quantity? (Ignore any uncertainty in the stride length while answering this.)

6 Conclusions

Here, we've looked at one attempt to model speed. Understanding this model led us to an equation for speed, distance, and time, and to the notion of dot diagrams. As we continue solving problems and building new models in class, we'll introduce new technical concepts to deal with those problems and models, including new mathematical notation, creating and analyzing graphs, and distinguishing closely related concepts such as speed and *velocity*.

7 References

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<https://metro.co.uk/2009/10/14/prehistoric-aussie-aboriginals-could-have-outrun-usain-bolt-490193/>

8 Credits

footprint photograph, tracks photograph: Michael Amendolia, visitmungo.com.au/ancient-footprints

ticker tape photograph: Western Washington University, <https://physics.wvu.edu/ticker-tape-timer>

ticker tape diagram: Evan Toh, "Ticker Tape Timer", YouTube,
<https://www.youtube.com/watch?v=FcvW7PxqgTo>

9 Answers to exercises

2.1: The left foot is both in front of and slightly to the side of the right foot. If we measured right foot, left foot, right foot, etc. we'd measure a zigzag pattern, so instead we measure from right foot to right foot.

2.2: No. There is the same amount of time between each dot, so more dots means tracking the motion over a longer time. Object B was tracked for the most time and object A for the least time.

2.3: 3.0 s. Note that there are six dots, but only five gaps between the dots, so the answer is $0.6 \text{ s} \cdot 5 = 3.0 \text{ s}$. This is called a "fencepost" problem because a fence with six posts only has five sections.

2.4: No. If an object were to start from the right and move left, it would create an identical series of dots as an object starting from the left and moving right, so we can't tell from just the dots which way the object moved.

2.5: The slow-moving ball should have dots closer together, e.g.

slow • • • • • • • • • • • • • • • •

fast • • • • • •

2.6: The time between the dots being laid down.

2.7: Increase. The slower the ticker, the longer a time the car has to move between dots being made, so it will move further and the dots will be spaced further apart.

2.8: The same for both cars. If the ticker ticked different rates, then the spacing between the dots is not enough to compare the speeds. Only if the ticker was always going at the same rate are dot diagrams directly comparable.

2.9: The same rate all the time. If the ticker were to change its ticking rate, that would affect the dot spacing even without the cart changing speed. If both the tick rate and cart speed were changing at the same time, we'd have no way of knowing what combination of both was causing the pattern in the dots.

3.1: The speed is 0.005 m/s , or 0.5 cm/s . This is about one foot per minute. It's about the speed of a garden snail, or a slow-crawling sloth.

3.2: 0.65 s . Since $\text{distance} = \text{speed} \cdot \text{time}$, we can solve this for time to get $\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{2.08 \text{ m}}{3.2 \text{ m/s}} = 0.65 \text{ s}$.

This is a typical time between strides while running long distances.

3.3: 1.98 m . Using $\text{distance} = \text{speed} \cdot \text{time}$ we have $\text{distance} = (25 \text{ m/s}) \cdot (0.079 \text{ s}) = 1.98 \text{ m}$. Since this is how far the car moves with each revolution of the wheels, it is the circumference of the wheel.

3.4: Any numbers are acceptable, as long as the distance divided by time comes out the same for the two pairs of numbers you chose. You could start with any distance and time, then multiply them both by the same number to get another distance and time that works.

3.5: Object B. We can make up some numbers to understand this. Suppose object *A* has 2 m between dots and object *B* has 1 m . We could also let object *A* have 3 s between dots and object *B* have 1 s . Then object *A* travels at $\frac{2}{3} \text{ m/s}$ and object *B* travels at 1 m/s . Object *B* is faster.

3.6: 15 m/s . Going from *A*'s dot diagram to *B*'s, we cut the distance between dots in half, which cuts the speed in half. However, we cover that distance in one third the time, which multiplies the speed by 3. Starting with 10 m/s , if we cut it in half, then multiply by three, we get 15 m/s .

4.1: Some possibilities: Webb's measurement of the stride length wasn't perfect; different strides have slightly different lengths, and nothing can be measured perfectly. It isn't possible to determine someone's height perfectly from their footprint. Someone could have slightly larger or smaller feet compared to other people the same height as them; it's only a correlation. The model Webb used to go from height and stride length to time between strides must have uncertainty. Even people of the same height could use different stride lengths and time between strides while traveling the same speed. There's also no guarantee that the footprint hadn't eroded some or otherwise changed its dimensions in the 20,000 years since it was made.

4.2: We calculated Bolt's average speed over the entire 100 m race. Since he has to start stationary at the beginning of the race, it takes him a while to get up to his maximum speed, and his average speed is lower than his maximum. He also slows down a little bit near the end of the race due to fatigue, again reducing his average below his maximum.

5.1: 13.8% This is found from $\frac{1.0 \text{ m/s}}{7.2 \text{ m/s}}$ converted to a percentage.

5.2: The child. Their speeds are the same, but the child is much shorter, so speed divided by height is greater for the child.

5.3: $0.52 \pm 0.07 \text{ s}$, or $0.52 \pm 0.08 \text{ s}$. We can find the time using $\text{time} = \frac{\text{distance}}{\text{speed}}$ and plugging in Webb's measured stride length and Ruiz and Torices's estimated speed. To find the uncertainty, we can repeat the calculation with the approximate lower and upper bounds of Ruiz and Torices's speed estimate and notice the results are about 0.07 s or 0.08 s away from the best guess estimate.